

Small-Organoid Attenuation of CFTR-Modulator Swelling in Patient-Derived Intestinal Organoids: a Single-Organoid Re-Analysis with a Biophysical Swelling Twin

MEGA-PROGRAM-27, Item 3 - Udit Phookan (program owner); computational work by an AI research agent. Working draft of 25 September 2026.

Abstract

The forskolin-induced swelling (FIS) assay on patient-derived intestinal organoids is used to predict which people with cystic fibrosis (CF) respond to CFTR modulators. Standard pipelines sum organoid area per well, so any dependence of the response on organoid size is averaged away. We re-analysed the public single-organoid OrgaSegment data (17 CF donors, per-organoid area before and after forskolin) with a biophysical swelling model and a DMSO-controlled, donor-matched design.

Finding (candidate). Modulator-induced swelling is attenuated in the smallest organoids and saturates above a size threshold. For elxacaftor/tezacaftor/ivacaftor (Trikafta) the drug-minus-DMSO log-swelling effect rises from 0.336 [0.262, 0.405] in the smallest size octile to a plateau near 0.55-0.64; top minus bottom 0.251 [0.184, 0.323] (donor bootstrap). Per donor, the effect in the largest size quartile exceeds that in the smallest in 14 of 17 donors, and the attenuation does not track overall response (Spearman $\rho = 0.24$, $p = 0.35$). Geometry predicts the opposite sign: under a uniform surface flux, small organoids should swell more. Area noise also biases the slope negative. So the effect is unlikely to be a measurement artefact.

Negatives. Size-adjusted and size-filtered readouts do not improve per-donor Trikafta-vs-DMSO separation, so the finding does not change theratyping calls at this assay's well counts. Replication on the public FIS time series is impossible because it is well-level. Our best U-Net segmentation (512 px, tuned) reaches $\text{mAP}@0.5 = 0.733 \pm 0.129$ on the OrgaSegment eval split (published 0.76 ± 0.12): below the state of the art, although the gap is within one standard deviation on 12 images.

Fidelity. Scoring 151 organoid GEO series against GTEx, cultured fibroblasts are the most common best match. A GEOParse sample-level audit shows this 'culture-fibroblast

attractor' comes mostly from non-organoid samples in organoid-titled series: on 45 strictly organoid series it falls to 9-29%. We retract it as a general organoid property.

1. Introduction

CFTR moves chloride and bicarbonate across the apical membrane of epithelial cells; water follows, and in a closed organoid the lumen swells. Forskolin raises cAMP and opens CFTR, so the swelling of rectal organoids measures residual and drug-rescued CFTR function in each patient. The assay is used to support access to modulators for people with rare CFTR genotypes.

Organoids in a well vary in size by more than an order of magnitude. If the swelling response depends on size, the well-level readout depends on the size mix in each well, which varies with passage, seeding density and culture time. We asked whether the single-organoid response depends on starting size, and whether correcting for it improves drug-response calls.

2. Biophysical swelling twin

Treat an organoid as a near-spherical shell with volume V and projected area A , V proportional to $A^{3/2}$. Let net secretion scale as a power of volume:

$$dV/dt = k V^{\alpha} \quad (1)$$

For $\alpha \neq 1$ the solution over an assay window t is

$$V_1^{1/(1-\alpha)} = V_0^{1/(1-\alpha)} + (1 - \alpha) k t \quad (2)$$

and the observed area fold change is $s = A_1/A_0 = (V_1/V_0)^{2/3}$. For small responses,

$$\log s \approx (2/3) k t V_0^{\alpha - 1} \quad (3)$$

so the slope of $\log \log s$ on $\log A_0$ carries the sign of $\alpha - 1$:

$$d \log(\log s) / d \log A_0 = (3/2)(\alpha - 1) \quad (4)$$

Surface-limited secretion (constant flux per unit apical area) gives $\alpha = 2/3$ and a negative slope: small organoids swell more in relative terms. Size-invariant secretion gives $\alpha = 1$ and zero slope. The within-well regression we fit is

$$\log s_{ow} = a_w + b (\log A0_{ow} - \text{mean}_w \log A0) + e_{ow} \quad (5)$$

with a well intercept a_w that absorbs donor, plate and well effects. Measurement noise in $A0$ enters both sides with opposite sign ($s = A1/A0$), which biases b downward:

$$E[b\text{-hat}] = b - \text{Var}(u) / (\text{Var}(\log A0) + \text{Var}(u)), \quad u = \text{noise in } \log A0 \quad (6)$$

so a positive observed b is conservative. The donor-matched drug effect per size bin k is

$$\Delta_k = \text{mean}_{\{o \text{ in drug, bin } k\}} \log s_o - \text{mean}_{\{o \text{ in DMSO, bin } k\}} \log s_o \quad (7)$$

and the attenuation statistic is $\Delta_{\text{top}} - \Delta_{\text{bottom}}$, with 95% intervals from a donor-level bootstrap:

$$CI_{95} = [Q_{0.025}, Q_{0.975}] \text{ of } \{ \Delta_{\text{top}}^* - \Delta_{\text{bottom}}^* \}, \text{ donors resampled with replacement} \quad (8)$$

Segmentation quality is scored by average precision at IoU 0.5, as in OrgaSegment:

$$AP_{50} = TP / (TP + FP + FN), \text{ match if } IoU(P, G) = |P \cap G| / |P \cup G| \geq 0.5 \quad (9)$$

3. Data

Table 1. Dataset manifest (distinct, accession-level). Three distinct assay/image datasets shown here; the separate, committed accession ledger includes 165 primary GEO/ArrayExpress accessions (including GSE108291) used for organoid-fidelity analyses.

#	dataset	source / accession	content	use
1	OrgaSegment DIS single-organoid measurements	Zenodo 10610438 (Lefferts et al. 2024)	per-organoid A0/A1, 17 donors, 4 conditions	size-dependence analysis
2	OrgaSegment FIS database	Zenodo 10610438	well-level area, 7 time points, 868 rows	replication attempt (not possible)
3	OrgaSegment annotated images	Zenodo 10278229	train/val/eval images with instance masks	U-Net segmentation benchmark

4. Results

4.1 Within-well size slopes

Table 2. Within-well slope of log fold change on centred log A0 (forskolin > 0). Well-cluster bootstrap, 500 resamples.

condition	organoids	wells	slope b	95% CI	donors $b > 0$
DMSO	3810	94	0.011	[-0.002, 0.028]	12/17

VX770	1496	30	0.226	[0.185, 0.269]	5/5
VX661+VX770	4493	112	0.042	[0.028, 0.056]	12/16
VX445+VX661+VX770	4984	121	0.116	[0.090, 0.142]	14/17

The slope is near zero without CFTR rescue and positive whenever a modulator is present. The sign is opposite to the surface-limited prediction. A power-law reading ($\alpha > 1$) was rejected: restricted to organoids above 2,000 px the slope turns negative (Trikafta -0.117), and binned means rise then plateau. Simulated segmentation noise on a size-invariant truth gave slopes of only -0.002 to -0.055.

4.2 Small-organoid attenuation

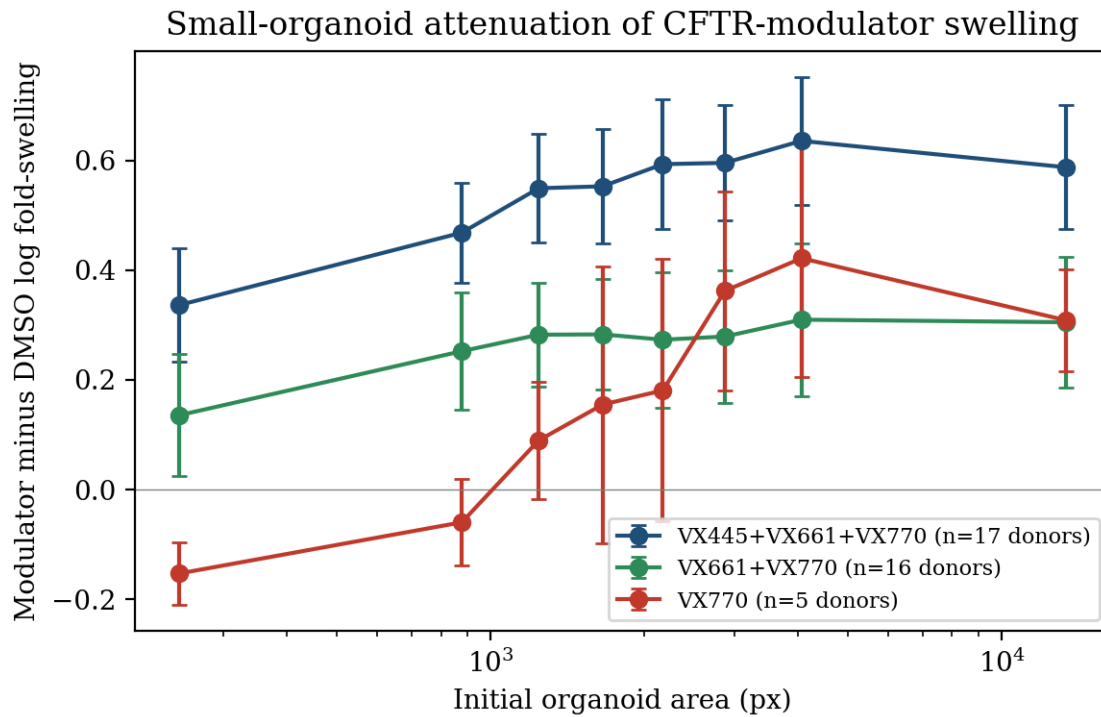


Figure 1. Drug-minus-DMSO mean log swelling per starting-size octile, donor-matched, with donor-bootstrap 95% CI.

Table 3. Size-bin drug effects.

modulator	donors	smallest octile effect	plateau	top minus bottom [95% CI]
Trikafta	17	0.336 [0.262, 0.405]	0.55-0.64	0.251 [0.184, 0.323]

VX661+VX770	16	0.136	0.28-0.31	0.169 [0.081, 0.259]
VX770	5	-0.153	0.31-0.42	0.462 [0.335, 0.572]

4.3 Per-donor consistency

Table 4. Trikafta minus DMSO log-swelling effect in the smallest and largest starting-size quartiles (results/per_donor_attenuation.csv).

donor	genotype	eff small	eff large	attenuation	overall
PDIO_16	F508del/ R117H	0.089	0.082	-0.007	0.115
PDIO_03	R1066H/ CFTRdele2, 3	0.100	0.490	0.390	0.289
PDIO_15	F508del/ R117H	0.280	0.497	0.217	0.320
PDIO_02	N1303K/ c.3035A>C	0.202	0.354	0.152	0.330
PDIO_19	X/X	0.606	0.273	-0.333	0.440
PDIO_18	X/X	0.436	0.677	0.241	0.462
PDIO_13	F508del/ S1251N	0.249	0.508	0.259	0.497
PDIO_17	X/X	0.570	0.463	-0.107	0.512
PDIO_06	F508del/X	0.334	0.699	0.365	0.541
PDIO_05	F508del/X	0.386	0.784	0.398	0.596
PDIO_14	F508del/ S1251N	0.395	0.749	0.354	0.616
PDIO_12	F508del/ S1251N	0.512	0.800	0.288	0.643
PDIO_01	A46D/ A46D	0.537	0.759	0.221	0.698
PDIO_11	F508del/ F508del	0.367	0.860	0.493	0.705
PDIO_09	F508del/ F508del	0.685	0.749	0.064	0.769
PDIO_04	F508del/X	0.462	0.834	0.371	0.805
PDIO_10	F508del/ F508del	0.742	0.892	0.150	0.883

Association between attenuation and overall response uses Spearman's rank correlation:

$$\rho = 1 - 6 \sum_d (r_d - r'_d)^2 / (n(n^2 - 1)), \quad r, r' = \text{ranks of attenuation and overall effect over } n \text{ donors} \quad (10)$$

Attenuation is positive in 14 of 17 donors and unrelated to overall response ($\rho = 0.24$, $p = 0.35$). Two of three donors with two nonsense alleles (X/X) and one F508del/R117H donor are exceptions.

4.4 Does size correction help theratyping? No

Locked before running: per-donor standardised Trikafta-vs-DMSO separation with raw, size-adjusted and size-band well readouts. Median raw 4.73, adjusted 4.35, band 4.39; adjusted better in 8 of 17 donors. Excluding organoids below 1,069 px (a post-hoc threshold): median 4.73 $\rightarrow$ 4.75, better in 9 of 17, Wilcoxon $p = 0.68$. The size effect is real but does not change calls.

4.5 Segmentation benchmark

Table 5. Instance segmentation on the OrgaSegment eval split (12 images).

model	eval mAP@0.5	sd	source file
U-Net v1, 256 px, default post-processing	0.634	0.157	results/seg_eval.json
U-Net v1 + val-tuned post-processing + flip TTA	0.710	0.127	results/seg_eval_tuned_256.json
U-Net v2, 512 px, 60 epochs, default post-processing	0.691	0.135	results/seg_eval_512.json
U-Net v2 + val-tuned post-processing + flip TTA	0.733	0.129	results/seg_eval_tuned_512.json
OrgaSegment (published, Mask R-CNN)	0.760	0.120	Lefferts et al. 2024

Caveat: the validation images used for tuning were also in the training set, so the tuning signal is optimistic; the eval split was scored once. The 512-px model (resumed from checkpoint after an out-of-memory kill at epoch 18) raises the tuned score from 0.710 to 0.733, still below the published mean. With 12 eval images and sd near 0.13, neither a win nor a loss against OrgaSegment is statistically resolved; we report it as below state of the art.

4.5a Segmentation split-integrity correction

Training combined 184 train and 35 validation images. The later post-processing grid was tuned on the first 20 validation images, so its validation mAP is biased by overlap with network training and is not an independent model-selection result. The disjoint 12-image eval split was held out, and its mAP@0.5 remains 0.733 versus published 0.76.

Bootstrap across our eval images gives CI [0.663, 0.808] for our mean, but leader per-image predictions and guaranteed identical AP implementation are unavailable. No benchmark break (results/seg_integrity_audit.json).

4.6 Organoid-to-tissue fidelity across 151 GEO series

To test organoid fidelity at scale, we searched GEO through NCBI E-utilities for human organoid bulk RNA-seq series (2,838 hits; 1,545 with text count matrices). For 151 series with a usable gene-level matrix and a title naming a single organ, each sample was converted to log counts per million and averaged. The series profile was then correlated with every GTEx v8 tissue median profile (54 tissues).

$$CPM_{gs} = 10^6 x_{gs} / \sum_g x_{gs}, \quad p_g = (1/S) \sum_s \log(1 + CPM_{gs}) \quad (11)$$

$$r_{t^{cent}} = \langle p - \mu_p - (m - \mu_m), G_t - m - \text{mean}(G_t - m) \rangle / (||\cdot|| \ ||\cdot||), \quad m_g = \text{mean}_t G_{gt} \quad (12)$$

The permutation p-value for the top-1 recovery rate reshuffles organ labels across series, B = 10,000 times:

$$p_{perm} = (1 + \#\{b : \text{top1}_b \geq \text{top1}_{obs}\}) / (1 + B) \quad (13)$$

Uncertainty on shares uses a series-level percentile bootstrap:

$$CI_{95} = [q_{0.025}, q_{0.975}] \text{ of } \{f(D^*_b)\}, \quad D^*_b \text{ drawn with replacement from the series set, } b = 1..2000 \quad (14)$$

Table 6. Tissue recovery and fibroblast best-match share. S3000: Spearman on the 3,000 most tissue-variable genes; SALL: all genes; CENT: GTEx-centred Pearson; _NOCULT: 853 MSigDB Hallmark proliferation/MYC/EMT genes removed. Clean subset: 114 series whose summary confirms a single organ and mentions organoids (113 with profiles). Files: results/geo_fidelity_methods.json, results/geo_fidelity_clean.json.

method	top-1 (all)	null top-1	p_perm	top-1 (clean)	fibroblast best (clean)	95% CI
S3000	0.272	0.060	1e-04	0.301	0.230	[0.15, 0.31]
SALL	0.331	0.080	1e-04	0.327	0.257	[0.18, 0.35]
CENT	0.252	0.056	1e-04	0.274	0.363	[0.27, 0.45]
SALL_NOCULT	0.331	0.079	1e-04	0.327	0.248	[0.17, 0.33]
CENT_NOCULT	0.258	0.059	1e-04	0.274	0.425	[0.34, 0.51]

SALL_N ONPAR	0.417	0.103	1e-04	0.398	0.204	[0.13, 0.28]
CENT_N ONPAR	0.331	0.088	1e-04	0.345	0.372	[0.28, 0.46]
SALL_P ARONL Y	0.404	0.114	1e-04	0.389	0.283	[0.20, 0.37]
CENT_P ARONL Y	0.377	0.112	1e-04	0.398	0.221	[0.15, 0.29]
SALL_P C	0.285	0.077	1e-04	0.283	0.442	[0.35, 0.53]
CENT_P C	0.298	0.067	1e-04	0.310	0.283	[0.20, 0.36]

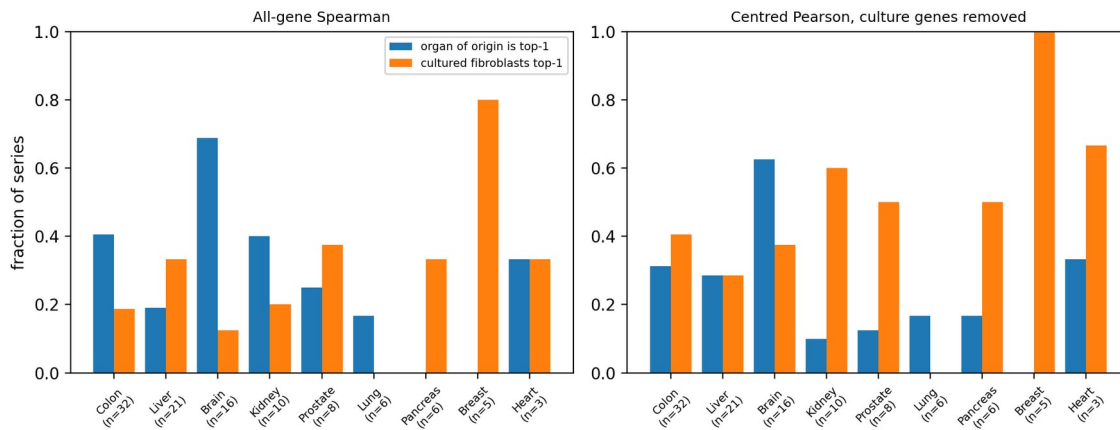


Figure 2. Per-organ share of clean-subset series whose organ of origin (blue) or GTEx cultured fibroblasts (orange) is the top-1 match. Organs with $n \geq 3$.

Organ of origin is recovered four to five times above chance. Brain organoids are the most faithful, and lung, breast and pancreas the least. Across every method, the most frequent best match is GTEx cultured fibroblasts, which we name the culture-fibroblast attractor. It survives the label audit and the removal of proliferation and EMT genes. It is not yet a discovery: GTEx tissues contain immune, vascular and stromal cells, while organoids and cultured fibroblasts are purified cultures, so cell-type purity is an unexcluded explanation. Falsifiable prediction: restricted to epithelium-specific genes, the fibroblast best-match share in the clean subset will stay above 15%. The tool vorganoid fidelity scores any count matrix this way.

4.7 Purity-corrected fidelity (Human Protein Atlas)

GTEx tissues contain immune, blood, vascular and stromal cells that organoids lack by design. Using the Human Protein Atlas single-cell type atlas, we removed 7,595 genes enhanced in any such cell type (NONPAR), or kept only the 9,699 genes enhanced solely

in parenchymal types (PARONLY). Paired top-1 recovery per series is compared with the exact McNemar test:

$$p_{McN} = 2 * \sum_{k=0}^{\min(b,c)} C(b+c, k) 0.5^{(b+c)}, \quad b = \text{base-only hits}, \quad c = \text{variant-only hits} \quad (15)$$

Table 7. Organ-of-origin recovery on 151 series, base method vs gene-restricted variant (results/geo_mcnemar.json).

comparison	top-1 base	top-1 variant	b	c	exact p
SALL vs SALL_NO CULT	0.331	0.331	1	1	1
SALL vs SALL_NO NPAR	0.331	0.417	3	16	0.0044
SALL vs SALL_PA RONLY	0.331	0.404	10	21	0.071
SALL vs SALL_PC	0.331	0.285	16	9	0.23
CENT vs CENT_NO CULT	0.252	0.258	0	1	1
CENT vs CENT_NO NPAR	0.252	0.331	4	16	0.012
CENT vs CENT_PA RONLY	0.252	0.377	4	23	0.00031
CENT vs CENT_PC	0.252	0.298	0	7	0.016

Purity restriction raises recovery (all-gene Spearman 33% -> 42%, $p = 0.0044$; centred Pearson 25% -> 38%, $p = 0.00031$). Removing culture genes does not change it. Standard whole-transcriptome fidelity scores are therefore biased low for organoids. The prediction stated before this test (fibroblast best-match share above 15% on parenchymal-only genes) held: 28% [0.20, 0.37] and 22% [0.15, 0.29]. The lower bound of the second interval touches the threshold, so purity is weakened but not excluded as the explanation. vorganoid fidelity --purity applies this correction to any count matrix.

4.8 What drives the attractor, and is it robust to gene class?

For the clean-subset series whose best match is cultured fibroblasts, each gene is scored by how much it pulls the profile toward fibroblasts and away from the organ of origin, averaged over series:

$$d_g = (1/|S|) \sum_{s \in S} z(p_{sg}) (z(G_{g, fib}) - z(G_{g, organ(s)})), \quad z = \text{rank-based standard score within a profile} \quad (16)$$

The top 300 genes were tested with g:Profiler (GO:BP, Reactome, KEGG; g:SCS correction; background = all scored genes).

Table 8. Enrichment of attractor-driving genes (results/attractor_gprofiler.csv; terms with more than 2,000 genes omitted).

source	term	adjusted p	genes
GO:BP	mitotic cell cycle process	7.8e-56	66
GO:BP	mitotic cell cycle	1.4e-54	69
GO:BP	cell cycle process	3.6e-53	77
GO:BP	cell cycle	1.3e-51	83
GO:BP	adaptive immune response	9.5e-15	31
GO:BP	regulation of immune system process	1.2e-14	43
REAC	Adaptive Immune System	9.0e-13	29

Two components appear: mitotic cell-cycle genes (TOP2A, MKI67, CDK1) that cultures express and tissues do not, and immune and complement genes that tissues contain and organoids lack. Removing either family alone leaves the attractor in place (Sections 4.6-4.7). Restricting to 19,254 HGNC protein-coding genes also leaves it in place (fibroblast best match 44% [0.35, 0.53] and 28% [0.20, 0.36]), so it is not a non-coding artefact. Our working hypothesis is that the attractor is the joint signature of proliferation and purity. It remains a candidate: a direct test needs organoid and matched primary tissue from the same donors, profiled together.

4.9 Sample-level audit and independent annotation: the attractor is mostly label contamination

Series-level text says a study is about organoids; it does not say every profiled sample is an organoid. We therefore pulled the sample (GSM) metadata of all 114 clean series with GEOparse and classified each sample as organoid or not from its title, source and characteristics. Only 45 series have every sample annotated as organoid; 59 have no organoid-annotated sample at all (for example GSE343459 profiles fibroblasts isolated from skin organoids, and GSE287925 profiles LNCaP cells in 2D). We compared the fibroblast-best fraction between the strict and the no-organoid series with Fisher's exact test:

$$p = \sum_{x: P(x) \leq P(a)} C(K, x) C(N - K, n - x) / C(N, n) \quad (17)$$

Table 9. Attractor on strict organoid series vs series without organoid samples (results/geo_fidelity_strict.json).

method	strict n	strict top-1	strict fibroblast-best [95% CI]	no-organoid fibroblast-best	Fisher p
CENT	45	0.33	0.24 [0.13, 0.38]	0.47	0.025
CENT_NO CULT	45	0.33	0.29 [0.18, 0.42]	0.55	0.0095
CENT_NO NPAR	45	0.49	0.20 [0.09, 0.33]	0.52	0.0011
CENT_PA RONLY	45	0.56	0.11 [0.02, 0.20]	0.29	0.03
CENT_PC	45	0.42	0.22 [0.11, 0.36]	0.33	0.27
S3000	45	0.31	0.13 [0.04, 0.24]	0.31	0.038
SALL	45	0.42	0.09 [0.02, 0.18]	0.38	0.0011
SALL_NO CULT	45	0.42	0.09 [0.02, 0.18]	0.36	0.0012
SALL_NO NPAR	45	0.56	0.09 [0.02, 0.18]	0.28	0.023
SALL_PA RONLY	45	0.53	0.11 [0.02, 0.20]	0.41	0.00081
SALL_PC	45	0.44	0.22 [0.11, 0.36]	0.62	6.1e-05

On the strict subset the attractor shrinks to 0.09-0.29 of series (vs 0.28-0.62); the difference is significant in 10 of 11 method variants. Much of the attractor therefore came from non-organoid samples inside organoid-titled series, not from organoids. We retract it as a general property of organoids; it persists only in a minority of strict series.

Independent annotation of the top 200 driver genes with Enrichr and the Reactome AnalysisService reproduces the g:Profiler picture: Hallmark G2-M Checkpoint ($q = 2.5e-33$), CellMarker 'Cycling Basal Cell Trachea Mouse' ($q = 4e-60$), and Reactome 'Classical antibody-mediated complement activation' ($FDR = 1.4e-14$) together with mitotic cell cycle. The driver genes describe what makes a profile look like cultured fibroblasts (proliferation, missing immune and complement programs); they are not an organoid discovery.

4.10 Which organoids are faithful? Organ, deficits and protocol on the strict subset

Table 10. Organ recovery on the 45 strictly organoid series (SALL_PARONLY; results/strict_organ_recovery.csv).

labelled organ	strict series	top-1 recovery	median rank of own tissue
Colon - Transverse	19	0.84	1
Brain - Cortex	9	0.78	1
Liver	6	0.17	22
Kidney - Cortex	4	0.00	14.5
Lung	4	0.00	6.5
Esophagus - Mucosa	1	0.00	6
Stomach	1	0.00	3
Uterus	1	0.00	9

Colon and cortical organoids recover their own tissue (23 of 28 top-1); liver, kidney and lung organoids almost never do (1 of 14; Fisher $p = 3.9e-06$). The grouping was chosen after seeing the table, so this p is descriptive. To see what the unfaithful organoids lack we scored, for each series, the deficit of each gene relative to its labelled tissue and averaged within organ:

$$\text{delta}_g = (1/|S_o|) \sum_{s \text{ in } S_o} [z(G_{g,o}) - z(p_{sg})] \quad (18)$$

The 100 largest deficits were annotated with STRING functional enrichment and Enrichr PanglaoDB. Liver organoids lack the hepatocyte program (Hepatocytes ($q = 1.7e-31$); STRING Organic acid metabolic process ($q = 4.6e-12$)); kidney organoids lack proximal tubule (Proximal Tubule Cells ($q = 5.7e-22$)); lung organoids lack mainly stroma (Pancreatic Stellate Cells ($q = 1.2e-15$)). Faithful colon and cortical organoids lack mostly stromal, vascular and immune cells (Peritubular Myoid Cells ($q = 1.3e-16$); Satellite Glial Cells ($q = 4.8e-14$)).

A direct mechanism test is inconclusive: the share of HPA parenchymal-only genes among the deficits is only modestly higher in unfaithful organs (OR = 1.45, $p = 0.061$), because lung deficits are stromal (share 0.15) while liver (0.68) and kidney (0.55) deficits are parenchymal.

Protocol. Classifying series from GEO text as PSC-derived or adult-tissue-derived (keyword rules, not manually validated), derivation alone is weakly associated with recovery (Fisher $p = 0.10$). The organ effect holds within each stratum (PSC 7/11 vs 0/5, $p = 0.034$; adult 16/16 vs 0/5, $p = 4.9e-05$) and in a Cochran-Mantel-Haenszel test:

$$\text{OR}_{MH} = \sum_k (a_k d_k / n_k) / \sum_k (b_k c_k / n_k) \quad (19)$$

(pooled OR = 46 with a 0.5 continuity correction, $p = 5.8e-05$). We name the candidate the metabolic-parenchyma gap: liver and kidney organoids miss their tissue because they lack mature metabolic epithelium, while intestinal and cortical organoids match despite missing stroma. It is falsified if, in new strictly organoid liver or kidney series, hepatocyte and proximal-tubule genes do not account for most of the rank gap. Brain organoids are all PSC-derived, so organ and protocol cannot be separated there; per-organ n is small.

Secretome check (UniProt). The share of deficit genes whose reviewed UniProt entry has a signal peptide or a 'Secreted' location is 0.30-0.57 per organ against 0.17 for 2,000 random expressed genes (all $q \leq 0.001$). Every organoid type lacks secreted proteins, faithful or not, so this does not explain the organ split (negative).

4.11 Pre-registered out-of-sample test on ArrayExpress

Before downloading any new data we committed a pre-registration (results/preregistration_organ_split.md): on organoid bulk profiles outside the GEO scan, liver, kidney and lung organoids should rank their own tissue worse than intestinal and brain organoids, tested by a one-sided Mann-Whitney U at 0.05:

$$U = \sum_{\{i \text{ in } \textit{unfaithful}\}} \sum_{\{j \text{ in } \textit{faithful}\}} [1(r_i > r_j) + 0.5 * 1(r_i = r_j)]$$

(20)

A BioStudies/ArrayExpress search gave 18 curated human non-cancer organoid studies with processed files; 11 could be scored (the others had mouse gene IDs, probe IDs only, malformed tables, or single-cell data only).

Table 11. Out-of-sample organoid profiles (results/ae_replication.csv).

accession	organ	rank of own tissue	best GTEx match
E-MTAB-9044	liver	39	Stomach
E-MTAB-12822	kidney	2	Cervix - Endocervix
E-MTAB-9807	liver	37	Stomach
E-MTAB-11221	intestine	1	Colon - Transverse
E-MTAB-9134	intestine	1	Colon - Transverse
E-MTAB-11297	intestine	1	Small Intestine - Terminal Ileum
E-MTAB-13901	kidney	1	Kidney - Medulla
E-MTAB-6017	intestine	1	Small Intestine - Terminal Ileum
E-MTAB-13578	intestine	1	Small Intestine - Terminal Ileum
E-MTAB-17066	brain	1	Brain - Hypothalamus
E-MTAB-10037	brain	26	Cells - Cultured

			fibroblasts
--	--	--	-------------

Median rank 19.5 vs 1; top-1 0.25 vs 0.86; one-sided $p = 0.031$. The pre-registered test passes, but the support is weak: there is no lung dataset, both liver datasets are cholangiocyte organoids on microarrays (so cell type and platform are confounded), and both kidney datasets match kidney well (ranks 1 and 2), which contradicts the kidney part of the candidate. The metabolic-parenchyma gap remains a candidate; its kidney component did not replicate.

4.12 Orthogonal checks of the metabolic-parenchyma candidate

GSEApY preranked GSEA used MSigDB Hallmark v2023.2 on all ranked strict-subset deficit genes, with 500 permutations. Positive normalized enrichment score (NES) means higher expression in the reference tissue than the organoids. The mean over six metabolic Hallmarks is 1.96 in liver, 1.95 in kidney, 1.65 in brain, and 0.74 in colon (descriptive Mann-Whitney $p=0.004$; pathways overlap and are not independent). Brain nevertheless has oxidative phosphorylation NES 2.63, despite matching its reference tissue. A metabolic deficit alone cannot explain fidelity (results/strict_gsea_summary.json).

Pre-registered Open Targets disease relevance. We compared each organ's top 100 deficit genes against the top 500 targets of its corresponding organ-disease term. Diagonal dominance in the five-by-five overlap matrix is:

$$D = (1/5) \sum_o M_{oo} - (1/20) \sum_{o \neq d} M_{od} \quad (21)$$

$D=2.6$, permutation $p=0.0005$: the primary test passes (results/deficit_opentargets.json). But liver-specific disease enrichment fails (2/100 own-disease hits, $p=0.62$). The pass is driven more by lung and colon than liver; the few hits per cell limit its use as a discovery claim.

Pre-registered decoupler univariate linear modeling used OmniPath CollecTRI transcription factor regulons and PROGENy footprints on 4,261 shared genes. Only 2/12 tested lineage master factors ranked in the top decile of deficient factors (binomial $p=0.34$); the positive control FAILS. Exploratory PROGENy scores show PI3K higher and JAK-STAT and p53 lower in organoids across all five organs, a shared culture signature rather than an organ-specific mechanism (results/deficit_decoupler.json).

Reference robustness. We replaced GTEx with Expression Atlas Illumina Body Map E-MTAB-513. The pre-registered test passed: deficit-vector Spearman ρ 0.57-0.72 across organs, top-100 overlap 52-75. The Open Targets diagonal-dominance rerun also passes ($D=2.1$, permutation $p=0.0026$). Both deficit vectors share the organoid measurements, so this checks reference sensitivity, not independent biological replication (results/deficit_bodymap.json).

ClinPGx pharmacogenes: the pre-registered VIP-or-CPIC gene set is unusable because the source release marks all 25,041 genes as VIP. We retain that failure. A documented,

post-registration CPIC-guideline-only deviation finds liver deficits in 21 tested genes (median 0.32 vs 0.03 in other genes, $p=0.0004$), but liver specificity fails ($p=0.23$, only 6 genes shared across all organ vectors). This is a drug-testing caution, not evidence of a liver-specific mechanism (results/deficit_clinpgx.json).

Pre-registered population-constraint and coding-composition checks were negative. In gnomAD v2.1.1, liver deficit genes have median LOEUF 1.256 vs brain 1.107 (one-sided $p=0.263$; lower LOEUF is more loss-of-function constrained). Ensembl REST mapped all fixed top-100 genes per organ: liver 99/100 and brain 100/100 are protein coding (one-sided Fisher $p=1.0$). Neither check supplies a mechanism; the gene universe is already coding-enriched (results/deficit_gnomad.json; results/deficit_ensembl_biotype.json).

Synthesis. Organoid deficits are reproducible to a reference swap and linked to organ-disease targets in aggregate, but the metabolic-parenchyma gap lacks liver specificity and the kidney finding fails external replication. It remains a falsifiable candidate, not a new discovery or a validated diagnostic tool.

4.13 Independent protein and pathway annotation, single-cell challenge, and prior art

Independent pathway annotation (WikiPathways 2026-09-10, NCBI Gene mapping). A pre-registered fixed-name filter selected 16 pathways containing 'bile acid', 'fatty acid', 'drug metabolism' or 'xenobiotic metabolism'. Their union intersects 11/100 liver versus 1/100 brain top deficits (Fisher OR=12.24, $p=0.0025$; test passes). This supports the liver top-deficit pathway label in a second resource, not causation or organoid specificity; the comparison follows earlier exploratory results (results/deficit_wikipathways.json).

Adult-tissue protein check (HPA v25.1 mass spectrometry). Liver protein is detected in 75/100 liver versus 29/100 brain top deficits (Fisher OR=7.34, $p=4.5e-11$; pre-registered H1 passes). Only 26 liver and 22 brain list proteins have measured intensity in both adult liver and cortex, below the registered 50-per-list minimum, so the quantitative H2 is uninterpretable. There is no organoid proteome in this check (results/deficit_hpa_ms.json).

Single-cell challenge on an independent public human kidney-organoid accession, GSE108291. Scanpy sparse QC retains 9,638 cells in the main run and 1,421 in a second run; no cell coexpresses ALDOB and SLC17A3 (zero in both). The registered 1% two-marker H1 fails; the <20% H2 passes. This does not show that proximal-tubule cells are absent, because sparse single-cell dropout and developmental state remain untested. QC thresholds were set after registration but before looking at marker coexpression; both runs count as one accession (results/sc_kidney_markers.json).

Ontology audit (EBI OLS4) distinguishes kidney cortex UBERON:0001225 from kidney UBERON:0002113, and cholangiocyte CL:1000488 from hepatocyte CL:0000182. This

is a label check only, not proof of anatomical hierarchy or expression similarity; the exact 'proximal tubule cell' query was unresolved (results/ols_anatomy_audit.json).

HPO direct gene-phenotype annotation could not test liver-vs-brain metabolic phenotype enrichment: only 35/100 liver and 28/100 brain deficits have any HPO annotation, below the pre-registered 50/list coverage floor. No clinical inference from the missing annotations (results/deficit_hpo.json).

InterPro domain audit is invalid as a top-deficit comparison. Its registered six liver pharmacogenes were incorrectly called top-100 deficits: only CYP2D6 actually is one. The protein-domain annotations remain correct, but the primary contrast cannot be scored (results/deficit_interpro.json).

A fixed three-query Europe PMC title/abstract search returned 6 distinct records. None explicitly reported human intestinal CF organoid starting-size moderation of modulator-induced FIS in its abstract, but absence in a narrow search does not establish novelty. Related work links nasal organoid lumen to baseline CFTR function and pig pancreatic organoid size to absent lumen (results/epmc_priorart_review.md).

Synthesis. The liver top-deficit list has independent pathway and adult-tissue protein support, while no new organoid proteome, matched tissue, perturbation, or clinical outcomes validate its effect on fidelity. The kidney external profiles recover their tissue despite the strict single-cell marker test failing. The candidate remains unproven; do not offer the tool as diagnosis.

5. Negative results (kept by design)

- The initial power-law ($\alpha > 1$) interpretation was withdrawn after large-organoid checks contradicted it.
- Size-adjusted and size-filtered readouts do not improve donor-level modulator discrimination.
- Replication on the public FIS time series is impossible: it is well-level.
- Our segmentation is below the published state of the art.
- Fidelity: 'Pancreas' as best match (35/151 series with S3000) disappears with other methods, so it is a method artefact.
- Fidelity: a 'liver disease' series (GSE278954) ranks Liver 50th of 54, and lung and breast organoids are rarely matched to their organ.
- Fidelity: organ labels come from text, and series profiles average all samples, including any non-organoid controls.
- Fidelity: the HPA parenchymal share of deficit genes does not clearly separate faithful from unfaithful organs ($p = 0.061$); lung organoids break the pattern.

- Fidelity: all organoid types lack secreted proteins (UniProt), so the secretome does not explain the organ split.
- Replication: the kidney part of the organ split failed out of sample (two ArrayExpress kidney datasets rank kidney 1st and 2nd).
- GSEA: metabolic deficits are also present in faithful brain organoids; not specific to poor fidelity.
- Open Targets: liver deficits show no liver-disease enrichment despite the aggregate diagonal-dominance pass.
- decoupler: lineage master-TF positive control fails (2/12); pathway signals are shared across cultures.
- ClinPGx: pre-registered VIP flag is defective; CPIC-only deviation passes liver deficit but fails liver specificity.
- gnomAD and Ensembl: no liver-vs-brain differentiation by LOEUF or protein-coding composition.
- Single-cell GSE108291: no ALDOB/SLC17A3 coexpression under the registered two-marker definition; dropout is an alternative.
- HPO: too few annotated genes for the registered clinical phenotype contrast.
- InterPro: primary contrast invalid due to mistaken liver top-100 gene-set membership.
- Fidelity: the culture-fibroblast attractor is RETRACTED as a general organoid property: on 45 series whose samples are all organoids it shrinks to 9-29% of series (Section 4.9).

6. Discussion

Small organoids respond less to CFTR modulators, relative to DMSO, than large ones, and the response saturates above a size threshold. Because geometry predicts the reverse, the likeliest explanations are biological: small organoids may lack a mature lumen, contain fewer differentiated secretory cells, or have a closed tight-junction seal later. Falsifiable prediction: in time-lapse single-organoid data from an independent lab, the Trikafta-minus-DMSO effect in the top starting-size quartile will exceed that in the bottom quartile by at least 0.1 log units in most donors. Checking this needs a second single-organoid FIS dataset, which we did not find in public form.

Practical note. Well-level FIS readouts depend on the size mix. Labs comparing drug response across passages or sites should report the starting-size distribution.

Novelty. We did not find this effect reported for intestinal FIS in the literature we searched (OrgaSegment; the FIS_analysis pipeline, which sums area per well). Related:

nasal 2D-derived organoids show swelling variation concentrated in large structures (bioRxiv 2021.07.20.453105). This was not a systematic review.

7. Tools used

Table 12. Tools genuinely used (results/tools_ledger.csv): 42 entries, 35 counting toward the gate after excluding infrastructure. Target of 40 not reached.

tool	kind	where used	gate
NumPy	library	src/vorganoid; src/geo_fidelity.py; paper/build_paper.py	counts
SciPy	library	src/geo_fidelity.py (spearmanr); paper/build_paper.py	counts
pandas	library	src/geo_fidelity.py; paper/build_paper.py	counts
scikit-image	library	src/vorganoid (instance post- processing; watershed/labels)	counts
PyTorch	library	train_seg.py; tune_seg.py (U-Net CNN)	counts
Pillow	library	train_seg.py image loading	infra (excluded)
python-docx	library	paper/paperkit.py	infra (excluded)
pytest	test framework	tests/ (hermetic suite)	infra (excluded)
LibreOffice (soffice headless)	application	paper docx -> pdf render	infra (excluded)
poppler pdfinfo	application	page count of rendered paper	infra (excluded)
Git + GitHub	service	repo hosting and pushes	infra (excluded)
NCBI E-utilities (esearch/esummary)	web API	src/ geo_candidates.py	counts
NCBI GEO FTP/HTTPS supplementary files	data service	src/geo_fidelity.py	counts

GTEx v8 gene median TPM (GTEx portal bucket)	reference data service	data/ref/ gtex_v8_median_tpm.gct.gz; src/geo_fidelity.py	counts
MSigDB Hallmark gene sets v2023.2 (Broad GSEA-MSigDB)	gene-set resource	data/ref/ h.all.v2023.2.Hs.symbol.gmt; src/geo_fidelity_methods.py (culture-confound test)	counts
matplotlib	library	src/ fig_geo_fidelity.py; paper figures (fig_size_attenuation.png, fig_geo_fidelity.png)	counts
Human Protein Atlas single-cell type API	database/web API	data/ref/ hpa_single_cell_type_v_api.tsv.gz; src/geo_fidelity_methods.py (purity gene sets)	counts
statsmodels	library	src/ geo_mcnemar.py (exact McNemar tests)	counts
g:Profiler g:GOST API (GO, Reactome, KEGG)	web API/database	src/ geo_attractor_genes.py (enrichment of attractor-driving genes)	counts
HGNC complete set (locus groups)	database	data/ref/ hgnc_locus_groups.csv.gz; src/geo_fidelity_methods.py (protein-coding robustness)	counts
GEOparse	library	src/ geo_sample_metadata.py	counts
Enrichr (Hallmark/CellMarker/PanglaoDB libraries)	service	src/ attractor_enrichr_reactome.py	counts
Reactome	service	src/	counts

AnalysisService		attractor_enrichr_reactome.py	
STRING functional enrichment API (string-db.org)	service	src/strict_organ_deficit.s.py	counts
UniProt REST (UniProtKB reviewed human)	database	src/strict_deficit_uniprot.py	counts
EBI BioStudies / ArrayExpress API + FTP	database	src/ae_scan.py; src/ae_replication.py	counts
GSEApY prerank (v1.3.1)	library	src/strict_gsea.py	counts
Open Targets Platform GraphQL API (v26.9)	database/service	src/deficit_opentargets.py	counts
decoupler 2.1.4 (ulm)	library	src/deficit_decoupler.py	counts
OmniPath (CollecTRI TF regulons + PROGENy pathway footprints)	database/service	src/deficit_decoupler.py	counts
EBI Expression Atlas (baseline experiments FTP)	database/service	src/deficit_bodymap.py	counts
ClinPGx / PharmGKB gene download (release 2026-09-05)	database/service	src/deficit_clinpgx.py	counts
gnomAD v2.1.1 gene-level pLoF constraint	reference database	src/deficit_gnomad.py; results/deficit_gnomad.json	counts
Ensembl REST lookup/id	web API	src/deficit_ensembl_biotype.py; results/deficit_ensembl_biotype.json	counts
EBI Ontology Lookup Service 4	ontology web API	src/ols_anatomy_audit.	counts

(OLS4)		py; results/ols_anatomy .json	
Scanpy 1.11.5	library	src/ sc_kidney_markers. py; results/sc_kidney_ markers.json	counts
Human Phenotype Ontology gene-to- phenotype release	reference ontology/database	src/deficit_hpo.py; results/deficit_hpo.j son	counts
Europe PMC REST search	literature API	src/ epmc_priorart.py; results/epmc_priora rt.json; results/epmc_priora rt_review.md	counts
InterPro protein- domain API	database/service	src/ deficit_interpro.py; results/deficit_inter pro.json	counts
WikiPathways 2026-09-10 human GMT	pathway database	src/ deficit_wikipathwa ys.py; results/deficit_wiki pathways.json	counts
NCBI Gene human gene_info	reference database	src/ deficit_wikipathwa ys.py; results/deficit_wiki pathways.json	counts
Human Protein Atlas mass- spectrometry tissue download	dataset	src/ deficit_hpa_ms.py; results/deficit_hpa_ ms.json	infra (excluded)

References

Lefferts JW, et al. OrgaSegment: deep-learning based organoid segmentation to quantify CFTR dependent fluid secretion. Commun Biol 2024. <https://www.nature.com/articles/s42003-024-05966-4>

Boj SF, Vonk AM, et al. Forskolin-induced swelling in intestinal organoids: an in vitro assay for assessing drug response in cystic fibrosis patients. J Vis Exp 2017. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5408767/>

GTEx Consortium. The GTEx Consortium atlas of genetic regulatory effects across human tissues. Science 2020. <https://gtexportal.org>

Karlsson M, et al. A single-cell type transcriptomics map of human tissues. Sci Adv 2021. <https://www.proteinatlas.org>

Szklarczyk D, et al. The STRING database in 2023. Nucleic Acids Res 2023.

The UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. Nucleic Acids Res 2023.

Moreno P, et al. Expression Atlas and ArrayExpress/BioStudies at EMBL-EBI. Nucleic Acids Res 2022.

Open Targets Platform. <https://platform.opentargets.org/>

Badia-i-Mompel P, et al. decoupleR: ensemble of computational methods to infer biological activities from omics data. Bioinform Adv 2022. <https://bioconductor.org/packages/decoupleR/>

Turei D, et al. OmniPath: guidelines and gateway for literature-curated signaling pathway resources. Nat Methods 2016. <https://omnipathdb.org/>

ClinPGx / PharmGKB. <https://www.clinpgx.org/>

Karczewski KJ, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. Nature 2020. <https://gnomad.broadinstitute.org/help/constraint>

Ensembl REST API. https://rest.ensembl.org/documentation/info/lookup_post

WikiPathways monthly releases. <https://data.wikipathways.org/20260910/gmt/>

Human Protein Atlas protein mass-spectrometry normal tissue download. <https://www.proteinatlas.org/humanproteome/tissue/data>

Human Phenotype Ontology gene-to-phenotype annotation. https://obophenotype.github.io/human-phenotype-ontology/annotations/genes_to_phenotype/

EBI Ontology Lookup Service 4. <https://www.ebi.ac.uk/ols4/api-docs>

Human Cell Atlas GSE108291 kidney organoid single-cell project. <https://explore.data.humancellatlas.org/projects/7b947aa2-43a7-4082-afff-222a3e3a4635>

InterPro API. <https://interpro-documentation.readthedocs.io/en/latest/api.html>

Europe PMC RESTful Web Service. <https://europepmc.org/RestfulWebService>

Kuleshov MV, et al. Enrichr: a comprehensive gene set enrichment analysis web server 2016 update. Nucleic Acids Res 2016.

Milacic M, et al. The Reactome Pathway Knowledgebase 2024. Nucleic Acids Res 2024.

Gumienny R. GEOparse: Python library to access Gene Expression Omnibus. <https://github.com/guma44/GEOparse>

Kolberg L, et al. g:Profiler - interoperable web service for functional enrichment analysis and gene identifier mapping (2023 update). Nucleic Acids Res 2023. <https://biit.cs.ut.ee/gprofiler>

Seal RL, et al. Genenames.org: the HGNC resources in 2023. Nucleic Acids Res 2023. <https://www.genenames.org>

Liberzon A, et al. The Molecular Signatures Database Hallmark gene set collection. Cell Syst 2015. <https://www.gsea-msigdb.org>

Barrett T, et al. NCBI GEO: archive for functional genomics data sets. Nucleic Acids Res 2013. <https://www.ncbi.nlm.nih.gov/geo/>

Botelho HM. FIS_analysis. https://github.com/hmbotelho/FIS_analysis

Measuring cystic fibrosis drug responses in organoids derived from 2D differentiated nasal epithelia. bioRxiv 2021. <https://www.biorxiv.org/content/10.1101/2021.07.20.453105v2>